Aditya Rajesh Sawant

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Experience

Safety Knights

Software Development Intern — *August 2026 — Present*

- Consolidated 5 sales and marketing tools (Instantly, Apollo, Metricool, Moosend, SendPilot) into one Next.js/TypeScript/Postgres CRM, replacing 5 separate logins and manual exports for a 3-person team
- Built a multi-source automation engine (webhooks, polling connectors with historical backfill, chained-step rules exposing 28 Instantly actions) with a visual rules editor, so non-engineers can build automations. Zero-click lead dedupe and sync across Apollo and Instantly cut ~12 hrs/week of CSV export and re-entry
- Shipped a GitHub Actions CI/CD pipeline that runs lint, type-checks and 1,000+ automated tests on every push, builds a Docker image tagged by commit SHA, and auto-deploys dev in about 5 minutes end to end, with prod promoted from the same tested image
- Moved hosting from Vercel and hosted Supabase to a self-hosted DigitalOcean droplet (Docker Compose, nginx, self-hosted Supabase) for full data control and fixed, predictable billing instead of usage-based pricing; added automated Drizzle migrations on every deploy and a BullMQ/Redis worker for background jobs

Software Development Intern — *January 2026 — May 2026*

- Architected and shipped a production EHS job board currently in beta with 4,000 users, spanning an Express/Node.js API (75+ endpoints, AES-256 encryption), an Angular v17 SSR consumer app, and a standalone admin console, a service-oriented architecture across 3 independently deployable codebases.
- Designed the data layer with 4 Mongoose models using compound, geospatial, and full-text indexes, automating hourly listing expiration via node-cron to keep job data fresh without manual intervention.
- Built a full candidate pipeline with GridFS document storage, 5-stage status tracking, 3 automated email flows, and multi-axis job search.

Research Assistant (Part-Time) — *February 2026 — Present*

SUNY Research Foundation, CUBS Lab (Prof. Srirangaraj Setlur)

- Built a fault-tolerant data ingestion pipeline scraping multiple federal and non-federal grant sources with a 10-worker async Python architecture that self-throttles to avoid overloading target sites.
- Designed pluggable source parsers and an LLM based parser behind a common interface with bounded back-pressure queues, keeping the 10-worker pool from overrunning downstream writes.
- Eliminated a cluster-internal auth-bypass by re-architecting Apache/Shibboleth SSO and Flask into a single-pod OpenShift sidecar (loopback-bound app, exec health probes, Secret-backed creds), making the API kernel-level unreachable except through SAML SSO.

Software Development Engineer — *August 2024 — June 2025*

CarWale [CarTrade Tech]

- Engineered and maintained the 'Oxygen Design System', a core library of 70+ reusable components in React and React Native powering front-end applications for 80 million monthly users, achieving a 10% bundle-size reduction on critical pages via a new counter-based hashing system.
- Led the architectural design of a dark theme for 60+ React Native components, using React Context and custom hooks to achieve seamless, no-reload theme switching.
- Cut a two-week estimate to 2 days by leveraging AI-driven automation and prompt engineering to update 250+ files.

Projects

Vantage — *August 2026 — Present*

Self-hosted TypeScript web analytics library with pluggable queue/store adapters and a client tracker (github.com/AdityaSawant0912/vantage)

- Architected a zero-infra-dependency TypeScript analytics core with pluggable QueueAdapter/StoreAdapter contracts, shipping Redis, BullMQ, and Postgres adapters that share one test suite; at-least-once delivery is effectively exactly-once by keying writes per tenant (source_id, event_id), and auth keys resolve once at ingest to isolate tenants
- Built a browser tracker with batched sendBeacon delivery, localStorage-backed offline retry, session and page-view chaining (prev_pv_id/seq), and IntersectionObserver impressions counting only active, focused, non-idle visible time

Optimus — *August 2026 — Present*

Zero-dependency TypeScript feature flag engine with React, Angular & SSR adapters (github.com/AdityaSawant0912/optimus)

- Designed a unified evaluation engine collapsing boolean, variant, and config flags into one core path, with deterministic id:flag:seed bucketing for consistent assignment across servers
- Built SSR support via single-evaluation snapshot hydration and published as @useoptimus/core (npm), with zero-dep architecture supporting Worker, Node, and browser runtimes

Skills

Programming Languages: JavaScript, TypeScript, Python, Rust, Java, C#

Frameworks: React, Node.js, Next.js, Angular, Redux

Databases: PostgreSQL, MongoDB, MySQL

Cloud & Infrastructure: Git, GitHub Actions, DigitalOcean, AWS, Docker, GCP, OpenShift, Elasticsearch

Education

Master of Science in Computer Science, University at Buffalo | GPA: 3.83/4 — *August 2025 — December 2026*

Computer Engineering, University of Mumbai | CGPA: 8.63/10 (~3.45/4.0 equivalent) — *August 2020 — June 2024*